

FARMINGTON, MO, USA

WMO: 724454

Lat: 37.761N

Lon: 90.428W

Elev: 289

StdP: 97.90

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 93996

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.7	-11.1	-19.0	0.7	-11.9	-17.1	0.9	-9.3	10.4	6.6	9.2	6.6	2.9	320	0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	34.0	24.4	32.7	24.3	31.6	24.0	26.2	31.5	25.5	30.9	24.8	29.8	3.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	20.5	29.5	23.8	19.4	28.7	22.9	18.3	27.5	83.0	31.8	79.5	31.1	76.6	29.8	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.9	7.1	-17.2	36.4	3.1	1.9	-19.5	37.7	-21.3	38.8	-23.0	39.9	-25.2	41.2		
(5)			-18.0	27.4	3.0	0.9	-20.1	28.0	-21.9	28.5	-23.5	29.0	-25.7	29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.2	0.7	2.8	7.8	13.2	18.3	23.0	25.0	24.3	20.0	14.0	7.2	2.0		
	DBStd	9.92	6.46	6.34	6.07	5.04	4.33	3.22	2.97	3.00	4.03	4.98	5.36	5.65		
	HDD10.0	1039	297	214	115	24	2	0	0	0	0	18	116	253		
	HDD18.3	2564	548	437	329	165	57	4	1	1	28	153	335	506		
	CDD10.0	2222	7	12	47	121	259	388	464	444	301	141	32	6		
	CDD18.3	704	0	0	3	12	56	143	206	187	79	17	1	0		
	CDH23.3	6590	0	1	18	102	455	1362	2058	1733	710	145	6	0		
	CDH26.7	2529	0	0	1	14	111	519	892	722	244	27	0	0		
(14) Wind	WSAvg	2.9	3.6	3.5	3.7	3.7	2.8	2.4	2.1	2.0	2.0	2.7	3.3	3.3		
(15) Precipitation	PrecAvg	1091	61	63	98	111	121	89	90	102	90	75	111	83		
	PrecMax	1661	177	132	232	313	332	316	259	306	310	180	300	242		
	PrecMin	687	2	2	20	31	34	16	5	12	9	2	14	14		
	PrecStd	218	42	31	51	61	66	57	51	60	67	48	73	50		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.6	21.6	26.0	28.7	31.3	34.1	37.1	36.3	33.7	29.1	23.7	18.7		
		MCWB	14.1	14.9	16.9	18.9	22.3	24.4	23.9	24.1	22.7	21.5	16.9	15.3		
	2%	DB	17.0	18.0	22.8	26.3	29.1	32.5	34.1	33.9	31.5	27.1	21.0	16.1		
		MCWB	13.3	13.0	15.4	17.6	21.6	24.2	24.9	24.0	23.1	19.5	15.2	12.7		
	5%	DB	13.6	15.9	20.2	24.0	27.7	31.3	32.7	32.4	29.4	24.8	17.9	13.0		
		MCWB	10.3	11.6	13.9	16.3	20.6	23.7	24.8	24.0	22.2	18.3	13.6	10.0		
	10%	DB	10.9	12.8	17.6	22.3	26.2	29.8	31.4	31.0	27.5	22.5	16.2	11.0		
		MCWB	7.9	8.7	12.5	15.7	19.8	23.1	24.4	23.6	20.6	17.0	12.0	8.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.9	16.8	18.5	20.6	24.1	26.3	27.5	27.0	25.1	22.9	18.7	15.9		
		MCDB	17.5	19.9	23.6	26.4	28.9	31.9	31.6	32.7	30.6	27.5	22.0	17.8		
	2%	WB	14.1	14.7	16.7	19.2	23.0	25.4	26.5	25.8	24.1	21.3	16.3	13.4		
		MCDB	16.2	17.4	21.1	24.1	27.6	30.9	31.5	31.2	29.2	25.1	18.9	15.5		
	5%	WB	10.9	12.2	15.2	17.9	21.8	24.6	25.7	25.0	23.1	19.7	14.7	10.9		
		MCDB	12.8	14.8	19.1	22.0	26.1	29.6	31.0	30.3	27.8	23.2	17.7	12.8		
	10%	WB	7.4	8.9	13.2	16.6	20.7	23.7	25.0	24.2	22.1	18.1	12.4	8.0		
		MCDB	10.4	12.7	17.2	20.8	24.9	28.3	30.2	29.2	26.0	21.2	15.6	9.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	10.4	11.4	12.1	11.2	11.2	10.9	11.3	12.2	12.0	10.5	9.0		
		MCDBR	13.4	14.8	14.2	14.1	12.6	12.7	12.4	13.1	13.9	13.8	13.3	12.0		
	5% WB	MCWBR	10.0	10.5	8.6	7.8	6.6	5.9	5.0	5.2	6.4	7.8	9.0	9.0		
		MCWBR	11.2	12.8	11.9	11.3	10.6	11.1	10.8	11.3	11.6	11.3	11.2	10.2		
(40) Clear-Sky Solar Irradiance	taub	taub	0.299	0.310	0.339	0.383	0.420	0.435	0.465	0.461	0.396	0.344	0.310	0.297		
		taud	2.507	2.462	2.438	2.311	2.257	2.236	2.179	2.189	2.352	2.494	2.534	2.521		
	Ebn at Noon	Ebn at Noon	892	924	925	897	865	848	819	813	852	873	872	868		
		Edh at Noon	78	93	105	127	136	139	146	142	114	90	76	73		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.18	2.82	3.83	4.99	5.61	6.39	6.15	5.57	4.76	3.52	2.41	1.82		
	RadStd	RadStd	0.24	0.32	0.36	0.31	0.49	0.43	0.40	0.30	0.29	0.38	0.26	0.19		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+82

Nomenclature: See separate page